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A survey of fake news detection: A comprehensive review of multimodal approaches

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A R T I C L E I N F O

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A B S T R A C T

With the rapid advancement of social media and artificial intelligence, fake news emerges as a critical societal challenge, contributing to widespread fake news and societal polarization. To address these challenges, multimodal approaches are considered promising solutions that integrate diverse data modalities to capture complementary information. This study provides a comprehensive examination of the evolution of fake news detection models, emphasizing the distinctive potential of multimodal techniques. We propose a novel taxonomy for categorizing multimodal approaches, which is structured around three critical dimensions: feature extraction, fusion methods, and learning objectives. Our review begins with a detailed comparison of key datasets, highlighting their characteristics, limitations, and applicability to multimodal tasks. We then conduct an in-depth analysis of multimodal detection techniques, exploring their methodologies and performance. In addition, we explore emerging technologies such as generative AI for robust detection and possible ethical considerations, including privacy risks and model biases. To address challenges such as cross-platform generalizability and real-world scalability, we propose several future directions and opportunities for developing effective multimodal fake news detection systems. Through this comprehensive review, we aim to contribute to the ongoing efforts to combat fake news by providing valuable insights and actionable guidance for researchers and practitioners in the field.

1. Introduction

In today's digital society, the rapid propagation of fake news has become a critical societal challenge that undermines public trust and exacerbates political polarization. The ubiquity of social media platforms, which a majority of Americans now rely on for news [1], has amplified the creation and spread of false information, making it increasingly difficult for individuals to distinguish between real and fake content [2,3].

The societal impacts of fake news are multifaceted, affecting critical domains such as politics [4], public health [5] and economic stability [6]. For instance, during the 2016 U.S. presidential election, numerous fake news stories circulated across social media platforms, with studies indicating their influence on voters' electoral decisions [7]. Beyond politics, fake news poses significant risks

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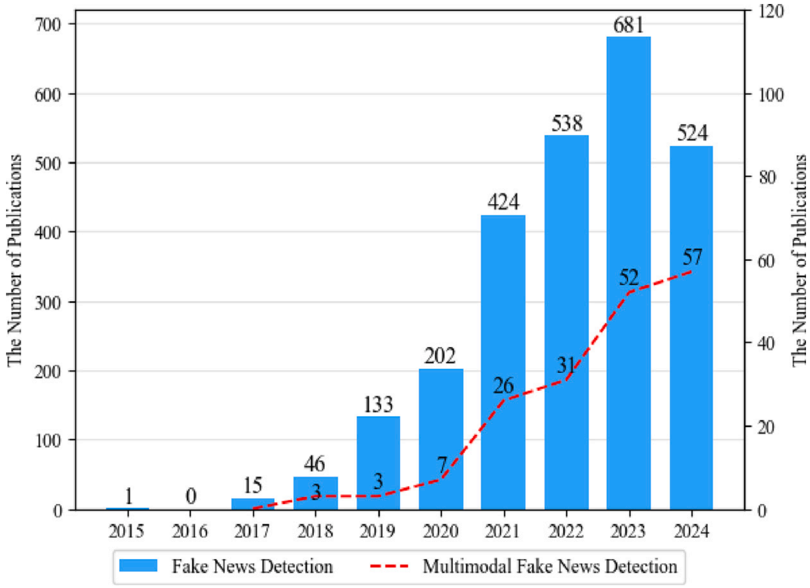


Fig. 1. Trend of fake news detection research from 2015 to 2024.

to public health. The COVID-19 pandemic was accompanied by an "*infodemic*" of false information regarding the virus and vaccine safety, which eroded public confidence in evidence-based health policies [8,9]. These examples underscore the profound societal threat that fake news poses across diverse sectors.

The proliferation of fake news emerged as a critical challenge, driving significant advancements and a growing body of research in fake news detection, as Fig. 1 shows the number of publications for fake news detection from 2015 to 2024. Early efforts in this domain primarily focused on analyzing single modalities, such as textual content [10] or visual elements [11,12] in isolation. However, such unimodal approaches are inherently limited, as they often fail to capture the complex interplay and inconsistencies between different data types that bad actors exploit to make fake news more persuasive and deceptive [13]. This limitation is becoming increasingly critical as modern social media platforms transition from text- and image-based content to video and audio formats [14,15].

In response to these limitations, multimodal approaches have emerged as robust solutions. Pioneering studies began integrating diverse data modalities, such as text and images, to capture complementary information and detect inconsistencies between them [16]. Subsequent advances in deep learning led to the development of powerful models, such as VisualBERT [17] and UNITER [18], which learn the correlations between textual and visual data to enhance detection accuracy. In addition to text and images, several researchers expanded to other modalities, such as audio [19] and social interaction information [20]. Studies have consistently demonstrated that multimodal frameworks outperform single-modality models. Consequently, research on multimodal fake news detection has rapidly expanded, a trend clearly illustrated by the growing number of publications in recent years, as shown in Fig. 1.

However, this rapid expansion led to a fragmented research landscape. Despite the proliferation of models, the field remains in its infancy, with notable research gaps and technical challenges that must be addressed. For instance, there is a significant lack of consensus on the relative advantages of various fusion strategies and optimal conditions for their application. Furthermore, many available datasets are limited to specific domains or languages, posing challenges in validating model performance in diverse real-world environments. Therefore, a systematic review is essential to organize this emerging body of work, identify key trends and limitations, and establish clear directions for future research.

Thus, this study aims to provide a comprehensive and structured analysis of the field by systematically examining key trends to guide future research. To achieve this, we introduce a novel taxonomy for categorizing multimodal approaches, which is structured around three critical dimensions: feature extraction, fusion methods, and learning objectives. Through a detailed comparison of key datasets and by applying our framework, we trace the development of detection models, provide a clear understanding of their underlying methodologies and critically evaluate their limitations. The goal is to offer valuable insights and actionable guidance for researchers and practitioners, ultimately contributing to the development of more robust and effective fake news detection systems.

The remainder of this paper is organized as follows. Section 2 reviews the prior literature and related surveys, identifying their limitations to establish the unique contribution of our comprehensive review. Section 3 provides a detailed analysis of the key datasets used in multimodal fake news detection, highlighting their attributes and limitations. Section 4 applies our proposed triaxial taxonomy to analyze and benchmark detection models across feature extraction, fusion strategies, and learning objectives. Finally, Section 5 discusses emerging challenges, such as AI-generated content and cross-platform generalizability, and outlines potential future research directions to advance the development of effective and scalable fake news detection systems.

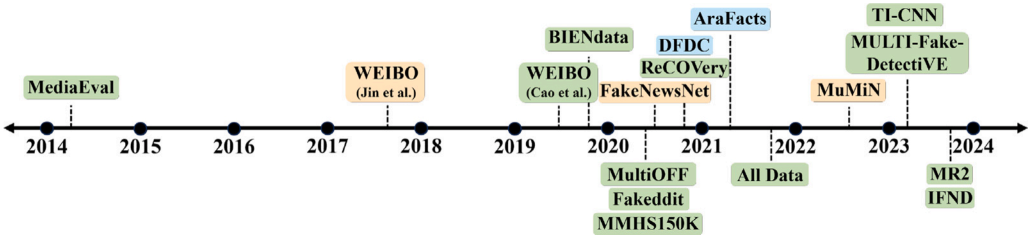


Fig. 2. Trends of multimodal fake news datasets; Green: Text + Image, Orange: Text + Image + social context metadata; Blue: Video. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2. Related works

Early studies on fake news detection naturally gravitated toward written text. Shu et al. [21] and the comprehensive synthesis of Zhou and Zafarani [22] established a foundational taxonomy identifying four primary dimensions for deception detection: stylistic features, knowledge-based cues, propagation patterns, and source credibility assessment. During this formative period, empirical investigations were constrained by limited data availability, with researchers predominantly utilizing small-scale monolingual English datasets such as LIAR [23]. These studies primarily employed support vector machines, early convolutional neural networks (CNNs), and long short-term memory (LSTM) classifiers.

Once larger datasets and transformer models [24] appeared, attention shifted to resources and end-to-end architectures. D'ulizia et al. [25] cataloged more than forty corpora, documenting how they were built and where bias might lurk, while Alam et al. [26] listed dozens of CNN, Bi-LSTM, and BERT variants. Although these reviews are undoubtedly valuable, they offer only brief references to multimodal research, entirely overlooking the audio and video modalities. Moreover, they fail to consider how the structure and composition of a dataset may constrain the types of fusion strategies that a model can pursue.

Recently, the conversation has shifted again, firmly toward multimodality. Nasser et al. [27] followed a Precise Surveys and Meta-Examinations (PRISMA) style pipeline, screening almost a thousand records and retaining 121 studies that span text, image, video, and audio. Zidan et al. [28] focused on text-image fusion and compared transformer cross-attention and multimodal GNNs while raising questions about interpretability.

Early efforts conceptualized fake news detection almost exclusively as a text-classification task. They clustered algorithms around lexical, stylistic, and psycholinguistic cues under a single-modality assumption and relied predominantly on English, politics-oriented corpora. Consequently, the interaction between textual evidence and social context signals, such as user graphs, remains largely unexamined.

The proliferation of image-rich platforms motivated a next survey waves that appended brief "vision" subsections to traditional text-centric methods. Most of these studies analyzed a simplistic early- and late-fusion, omitted audio and video modalities, and offered only superficial treatment of higher-order operators such as cross-modal attention or bilinear pooling.

Building on these partial attempts, a more comprehensive strand of multimodal surveys appeared. Recent multimodal surveys were conducted to cover text-image, text-video, and text-audio research. However, these surveys still evaluate models separately and mostly ignore issues such as training data biases, metadata, speech signals, and advanced fusion components.

To address these gaps, we advance the field in an integrated manner.

- We introduce a unified tri-axial map that situates all studies by feature extractions, fusion strategies, and learning objectives
- We examine numerous widely used fake news datasets, expose modality and language biases, and suggest guidelines for curating balanced multimodal benchmarks
- We explore how emerging generative models reshape detection challenges and raise privacy- and bias-related considerations

3. Datasets

Fake news detection emerges as a critical research domain in modern society, aiming to assess the authenticity of information. Fake news detection models developed to address this issue rely heavily on trustworthy datasets for training and evaluation, and the diversity and quality of these datasets significantly influence the outcomes of research. Because fake news can propagate through various modalities, such as text, images, and videos, the importance of datasets incorporating multimodal data is becoming increasingly prominent.

Fig. 2 illustrates the growth trend of multimodal datasets, encompassing text, image, and video modalities. The emergence of multimodal datasets began in 2014, with their gradual adoption and development in the early years. However, a significant surge in the creation of multimodal fake news datasets has been observed since 2020, coinciding with rapid advancements in deep and multimodal learning research. This period marks a pivotal shift in the exploration and application of multimodal approaches, particularly in fields such as fake news detection. Multimodal datasets play a critical role in enhancing the accuracy and robustness of detection systems, making them indispensable for addressing the complexities of misinformation in various contexts.

(a) Examples of fake news articles [29].

(b) Examples of fake news images [30].

Fig. 3. Examples of fake news benchmark dataset.

Fig. 3 shows examples of fake news articles and their corresponding image samples. Datasets for fake news detection can be broadly categorized into unimodal and multimodal datasets. Each type captures the unique characteristics of a specific modality while providing the diverse metadata required to detect and understand the spread of fake news. This section systematically categorizes the commonly used datasets for fake news detection based on modality and discusses their key features and applications.

3.1. Unimodal-based dataset

Unimodal datasets for fake news detection focus on a single modality, such as text or image. These datasets play a foundational role in understanding fake news characteristics. A summary of the representative unimodal datasets is presented in Table 1.

Text-based datasets, the most widely used, typically include news articles, social media posts, and headlines labeled as fake or real. They provide insights into the linguistic patterns, sentiments, and grammatical features commonly associated with fake news.

For example, the LIAR [23] dataset serves as a benchmark for evaluating the veracity of political statements. It includes 12,836 statements collected from PolitiFact[2] between 2007 and 2016, each annotated with one of six fine-grained labels: pants-fire, false, barely-true, half-true, mostly-true, and true. These categories make the LIAR dataset particularly well-suited for assessing text-based fake news detection models that require a fine-grained classification. In addition to the headlines and bodies of news articles, the dataset contains metadata such as the source of the statement, contextual information, and the profession of the person making the statement. This rich contextual data enables researchers to examine fake news within a broader sociopolitical context. The LIAR dataset is primarily used for detecting political fake news and supports multi-class classification tasks, offering a valuable resource for advancing text-based fake news detection methods.

The ISOT [31] dataset is another important resource for political fake news detection. It consists of 12,600 real news articles from Reuters and 12,600 fake news articles from Kaggle. The fake news samples were characterized by sensationalist headlines and misleading content, making the dataset ideal for studying the linguistic patterns and emotional cues that differentiate fake news from legitimate news.

The CREDBANK [32] dataset is a large-scale text-based resource designed for credibility analysis research. It contains 60 million global tweets collected between October 2014 and February 2015, spanning 1000 events. Each event is annotated with a credibility rating, allowing researchers to evaluate the reliability of the information shared on social media platforms.

The FacebookHoax [33] dataset focuses on scientific and conspiracy-related news collected from Facebook, distinguishing between fake and real news stories. It includes 8923 hoax posts and 6577 non-hoax posts, making it a valuable resource for analyzing the spread and characteristics of fake news in social media ecosystems. This dataset enables research into the linguistic and contextual features that differentiate hoaxes from legitimate content, laying the groundwork for developing models to mitigate the impact of fake news on social media.

The PHEME [41] dataset supports rumor detection research by providing Twitter data related to major social events. It includes tweets collected from five significant incidents: the Ferguson unrest (August 2014), Ottawa shooting (October 2014), Sydney hostage crisis (December 2014), Charlie Hebdo attack (January 2015), and Germanwings plane crash (March 2015). The dataset is categorized into rumors and non-rumors, containing 1972 rumor tweets and 3830 non-rumor tweets. Each tweet is also classified according to the specific event it references, making the dataset particularly valuable for analyzing the propagation and contextual characteristics of rumors.

[2] https://www.politifact.com/.

Table 1
Summary of unimodal fake news datasets.

Dataset	Description	Modality	Source	Labels	Data volume
2CTweets [34]	Tweets classification for incident-related events	Text	Twitter	Related, non-related	Not specified
BuzzFeed [35]	Hyperpartisan and fake news articles	Text	Facebook	Mostly true, mixture, mostly false, no factual content	1627
COVID-19 [36]	COVID-19 fake news dataset	Text	Public fact-verification sites, Social media	Real (22,916), Fake (19,728)	42,644
CREDBANK [32]	Social media credibility annotations corpus	Text	Global tweets	Credibility annotations	1000
CrisisLex [37]	Crisis-related lexicon for microblogs	Text	Twitter	Disaster-related (keyword-based: 9,777,964, location-base: 639,667)	10,417,631
Emergent [38]	Stance classification dataset for rumors	Text	Rumor sites, Twitter	For (1238), Against (394), Observing (963)	2595
FacebookHoax [33]	Hoaxes and non-hoaxes on social networks	Text	Facebook	Hoaxes (8923), Non-hoaxes (6577)	15,500
George McIntire Dataset [39]	Political fake news detection dataset	Text	Various news websites	Fake (3164), Real (3171)	6335
ISOT [31]	Benchmark dataset for fake news detection	Text	Reuters, Politifact, Kaggle	Fake (12,600), Real (12,600)	25,200
LIAR [23]	Truthfulness annotations on short statements	Text	PolitiFact	Pants-fire, False, Barely-true, Half-true, Mostly-true, True	12,836
MICC [40]	Image forensic dataset for copy-move detection	Image	Columbia Photographic Image Repository	Fake, Real	MICC-F220, MICC-F2000, MICC-F6000
PHEME [41]	Rumor detection dataset for major social events	Text	Twitter	Rumor (1972), Non-rumor (3830)	5802
Twitter Indian Dataset v3 [42]	Fake news detection in political and religious content	Image	Twitter	Fake (61), Valid (49)	110

While text-based analysis is essential, it often cannot account for visual deceptions that reinforce false narratives. Consequently, image-based datasets have been developed to analyze these visual cues. These datasets consist of manipulated or misleading visuals paired with annotations, allowing researchers to analyze visual cues such as inconsistencies, tampering, or contextual mismatches. Unimodal datasets not only aid in the development of modality-specific detection methods but also serve as a foundation for more sophisticated multimodal approaches by providing a deeper understanding of the unique attributes of each modality.

The MICC [40] dataset is a forensic images designed for the detection and analysis of copy-move forgeries. It consists of three subsets, each tailored to different scales and complexities in forgery detection. MICC-F220 contains 220 images with copy-move forgeries, MICC-F2000 includes 2000 images, and MICC-F6000 is a comprehensive set of 6000 high-resolution images with copy-move forgeries. This dataset is effective for learning the characteristics of image manipulation and evaluating the performance of algorithms designed for forgery detection. The MICC dataset is particularly suitable for testing traditional methods, such as SIFT-based clustering, and advanced deep learning techniques for detecting synthetic manipulations. Its diverse subsets make it a valuable resource for developing robust models capable of identifying and localizing image forgeries across various scenarios.

The Twitter Indian Dataset v3 [42] focuses on analyzing fake and authentic images shared on Twitter in India between November 2019 and November 2020, with a particular emphasis on political and religious events, which are common targets of fake news. The dataset consists of 110 images, of which 61 are labeled as fake and 49 as authentic. These images were collected by searching for morphed or forged news images on Indian fact-checking websites such as BoomLive, Alt News, and India Today, and then retrieving the corresponding tweets and news articles from Twitter. The dataset underwent a two-phase verification process: initially, images were cross-checked on fact-checking websites, and subsequently, they were manually annotated by peer reviewers who confirmed the authenticity of the images on the Twitter platform. This dataset provides valuable insights for studying the spread of fake images on social media and developing models aimed at detecting fake news.

3.2. Multimodal-based dataset

Unimodal datasets are essential for understanding the distinct characteristics of fake news within each modality. However, in real-world scenarios, fake news often involves multiple modalities such as text paired with images or videos. To address this complexity, multimodal datasets emerged as crucial resources, enabling the integration of diverse modalities to capture the rich and nuanced nature of fake news. By combining text, images, and videos with synchronized annotations, these datasets facilitate the analysis of cross-modal relationships and offer a more realistic representation of the fake news. This supports the development of advanced detection models that are capable of effectively handling multimodal inputs. Examples of multimodal datasets are presented in Table 2.

The DeepFake Detection Challenge (DFDC) [43] dataset is a large-scale resource developed by Facebook to advance deepfake detection research. The dataset comprises 104,500 manipulated and 23,654 authentic videos. The dataset includes synthetic videos generated using advanced deepfake techniques, featuring facial manipulation and audio alteration. It is widely used to evaluate the performance of video-manipulation-detection algorithms. By providing high-quality realistic synthetic videos, the dataset enables researchers to test their models under practical and challenging conditions. As such, it is a critical resource for developing robust solutions to combat video-based fake news and manipulations.

The AraFacts [45] dataset is a multimodal resource designed to detect fake news and fake news in the Arabic-speaking world. It integrates text, image, and video data, providing a comprehensive foundation for research in the domain. The dataset was constructed using data collected from various fact-checking organizations affiliated with the International Fact-Checking Network (IFCN). It spans several key topics, including politics, health, social issues, religion, and general sciences & arts. The politics category addresses fake news related to elections and political events, whereas the health category focuses on false information concerning medical and public health topics. The social category includes content related to societal issues and cultural narratives, and the religion category includes claims involving religious themes and controversies. Finally, general sciences & arts present broader topics in general knowledge and artistic themes.

The WEIBO [57] dataset includes both text and image data collected from the Chinese social media platform Weibo, categorized into rumors and nonrumors. It consists of 3749 rumor samples and 3783 non-rumor samples collected between May 2012 and November 2018. This dataset examines the interaction between textual and visual elements and explores how images can reinforce rumors related to specific topics. It is an essential resource for fake news detection research in the Chinese context, enabling integrated analyses of text and image modalities within multimodal-learning frameworks.

The MMHS150K [51] dataset is a multimodal collection of text and image data gathered from Twitter, designed to address tasks such as detecting hate speech. It is categorized into several classes, including non-hate, racist, and sexist content, reflecting the diversity of hate speech. MMHS150K provides researchers with an opportunity to study the intricate relationship between language and imagery in the context of harmful content online. Its multimodal nature enhances our understanding of how text and images contribute to the propagation of offensive messages, making it a valuable resource for developing and evaluating more robust detection models for harmful content.

The MediaEval [50] dataset includes both text and visual information related to major disasters, such as Hurricane Sandy (HS) and the Boston Marathon (BM). It comprises 10,757 fake text-image pairs, with 3540 real and 281 fake samples for HS and 460 real and 281 fake samples for BM. This dataset supports research focused on detecting false visual information disseminated through social media during disasters. This is particularly valuable for assessing the reliability of information in emergency contexts and addressing the challenges of fake news in critical situations.

The Multimodal Multilingual Fact-Checked Fake News (MuMIN) [55] dataset is a comprehensive collection of fact-checked, multilingual multimodal data gathered from social media platforms, designed to support fake news detection across a variety of topics, including COVID-19. It incorporates text, images, tweets, and user metadata, covering diverse languages and topics. The key components of the dataset include tweet text with corresponding fact-checking information, visual content linked to the tweets, and metadata about the tweet authors. MuMIN is well-suited for studying the interactions between text and images and analyzing the spread of fake news within specific user groups. This dataset enables a holistic approach to understanding and mitigating fake news in diverse social and linguistic contexts.

The FakeNewsNet [48] dataset is a widely used multimodal resource designed for fake news detection. It includes both text and image data, where the text consists of news article bodies and headlines, and the images are visual content associated with the news articles. In addition to textual and visual data, FakeNewsNet incorporates social context, enabling researchers to study how fake news spreads across social media platforms. The dataset is divided into two categories: political news sourced from PolitiFact and entertainment news from GossipCop. This division allows for a focused analysis of the textual and visual features associated with fake and real news within these specific domains. By integrating multimodal data, FakeNewsNet supports the exploration of correlations between text and images, making it a valuable tool for identifying the unique features of fake news and enhancing the development of detection models.

The Fakeddit [47] dataset is a multimodal resource designed to support fake news detection research and consists of both text and images collected from Reddit. With a total of 1,155,550 samples, including 628,501 fake news samples and 527,049 real news samples, Fakeddit enables sophisticated analyses that combine textual and visual data. This makes it particularly valuable for studies that leverage contextual cues from images to identify fake news. This dataset is widely used to explore textual and visual patterns associated with fake and real news, allowing researchers to evaluate the synergistic effects of multimodal data. Additionally, the integration of Reddit's unique social context provides an important resource for understanding the creation and dissemination of fake news, offering insights into how fake news propagates within online communities.

The MULTI-Fake-DetectiVE [53] dataset focuses on the Russia-Ukraine war and integrates text and images to support fake news detection and verification. It is structured around two subtasks: Subtask #1 classifies news as certainly fake, probably fake, probably real, or certainly real, whereas Subtask #2 evaluates the intent of posts, categorizing them as misleading, not misleading, or unrelated. This dataset is designed to assess the effectiveness of multimodal approaches, as combining text and images allows researchers to explore complex structures of fake news. MULTI-Fake-DetectiVE serves as a critical resource for developing and evaluating models that analyze the interplay between textual and visual elements in fake news and address the nuanced challenges of fake news detection and intent analysis.

Table 2
Summary of multimodal fake news datasets.

Dataset	Description	Modality	Source	Labels	Data volume
All Data [44]	Multimodal dataset for fake news detection with text and images	Text, Image	Various news articles and images	Real (8074), Fake (11,941)	20,015
AraFacts [45]	Arabic dataset of naturally occurring claims for fact-checking	Text, Image, Video	Fatabyyano, FactuelAFP Arabic, Misbar, Verify-Sy	False, Partly-False, Sarcasm, True	6,222
BIENdata [46]	Dataset for fake news detection from BIEN competition	Text, Image, OCR	News websites	Real (Text: 19,186, Image: 11,064, OCR: 5276), Fake (Text: 19,285, Image: 10,768, OCR: 3286)	68,865
DFDC [43]	DeepFake Detection Challenge dataset for deepfake video detection	Video	Synthesized videos	Real (23,654), Fake (104,500)	128,154
Fakeddit [47]	Multimodal dataset for fine-grained fake news detection on Reddit	Text, Image	Reddit posts	Real (527,049), Fake (628,501)	1,155,550
FakeNewsNet [48]	Repository with news content, social context, and spatiotemporal information	Text, Social context	Fact-checking websites	Real, Fake	Not specified
IFND [49]	Indian dataset for fake news detection with titles, images, and URLs	Text (Title, URL), Image (URL)	Indian news sites	Real (37,809), Fake (7271)	45,080
MediaEval [50]	Dataset focusing on computational verification in multimedia during crises	Text, Image, URL	Twitter	Real (Hurricane Sandy: 3540, Boston Marathon: 460), Fake (Hurricane Sandy: 10,757, Boston Marathon: 281)	15,038
MMHS150K [51]	Hate speech detection in multimodal tweets with images	Text, Image	Twitter	Non-Hate (112,845), Racist (11,925), Sexist (3495), Homophobic (3870), Religion-Based (163), Other (5811)	138,109
MR2 [52]	Multimodal retrieval-augmented rumor detection dataset	Text, Image	Twitter, Weibo	Rumor, Non-Rumor, Unverified Post	14,700
MULTI-Fake-DetectiVE [53]	Multimodal dataset for Russo-Ukrainian war fake news detection	Text, Image	Social media posts, news articles	Real (252), Fake (196), Misleading (485), Not Misleading (705)	1,638
MultiOFF [54]	Multimodal offensive content detection	Text, Image	Social media	Offensive (305), Non-offensive (438)	743
MuMiN [55]	Multilingual multimodal fake news dataset	Text, Tweet, Image, User	Fact-checking organizations, Twitter	Fake news, Factual, Other	984
ReCOVery [29]	Repository for COVID-19 news credibility research	Text, Image	Social media platforms (NewsGuard, Media Bias/Fact Check)	Reliable (1364), Unreliable (665)	2,029
TI-CNN [30]	Dataset for training CNNs on fake news detection	Text, Image	Megan Risda (240 websites), Authoritative news websites	Real (8074), Fake (11,941)	20,015
WEIBO [56]	Chinese dataset for fake news detection with text and images	Text, Image	Weibo posts	Real (Text: 19,186, Image: 20,461), Fake (Text: 19,285, Image: 13,635)	72,567
WEIBO [57]	Dataset for rumor detection using multimodal information	Tweet Text, Image, Social context	Weibo posts	Rumor (3749), Non-Rumor (3783)	7,532

3.3. Domain specific dataset

Moreover, domain-specific datasets are developed to support fake news detection, focusing on particular topics or fields. These datasets facilitate in-depth research into the characteristics and propagation mechanisms of fake news in specific domains. Notably, datasets centered on the COVID-19 pandemic and the U.S. presidential elections stand out as prominent examples. Such datasets are essential for understanding the unique structures and features of fake news in these areas and are invaluable for developing targeted and effective detection models. By incorporating these field-specific insights, domain-specific datasets enhance overall research effectiveness and expand the scope of fake news detection across diverse and critical real-world environments.

3.3.1. Covid-19

During the COVID-19 pandemic, fake news had a significant social impact worldwide. Datasets related to this crisis focus on analyzing and detecting false information concerning public health and infectious diseases. These resources provide a solid foundation for developing models that can address the challenges of fake news during health crises, ensuring accurate and reliable communication during emergencies.

The ReCOVery [29] dataset is a multimodal resource collected during the COVID-19 pandemic that comprises text, image, and video data. Based on news from reputable sources such as NewsGuard and Media Bias/Fact Check, this dataset offers a robust foundation for fake news research. This facilitates the study of the interplay between multiple modalities to detect and understand COVID-19-related fake news.

The COVID-19 Fake News [36] dataset, in contrast, is a text-based resource gathered from fact-checking platforms, such as PolitiFact, Snopes, and Boomlive. It focuses on analyzing the impact of fake news on public health during the pandemic and is widely utilized for developing models to curb the spread of fake news. This dataset is particularly valuable for examining the linguistic patterns and contextual features of fake news in textual form, supporting the creation of effective tools for detecting fake news in health-related contexts.

3.3.2. US election

The U.S. presidential election, especially the 2016 election, is one of the focal points of fake news research due to the widespread dissemination of fake news through social media platforms. Election-related datasets are instrumental in studying the generation and propagation mechanisms of political fake news articles.

The TI-CNN [30] dataset includes text and image data related to the 2016 U.S. presidential election, specifically designed for fake news detection. This dataset is a valuable resource for analyzing the interplay between textual and visual elements in political fake news and for evaluating multimodal detection models.

The BuzzFeed [35] dataset, which is based on political news collected from Facebook, focuses on understanding how political fake news spreads and gains credibility on social media. This is particularly useful for studying user interactions and reactions to fake news on platforms such as Facebook, providing insights into the behavioral dynamics of fake news dissemination.

The George McIntire [39] dataset includes text data related to the 2016 U.S. presidential election, including 3164 fake and 3171 real news articles. It is widely used to detect political fake news, supporting research on linguistic cues and emotional elements within textual data to better understand the characteristics of fake news in political contexts.

3.4. Taxonomy of misinformation

As multimodal fake news detection matures, the industry shifted from a simple binary of real versus fake to more nuanced labeling schemes. This transition is most evident in the construction of modern benchmarks that categorize misinformation based on intent and the nature of the content's distortion. As observed by Nakamura et al. [47] and Segura-Bedmar and Alonso-Bartolome [58], fake news can be categorized into several distinct technical and stylistic types, including satire or parody, which has no intention to cause harm but has the potential to fool, and misleading content, which involves the misleading use of information to frame an issue or individual. Other prominent categories include imposter content, where genuine sources are impersonated, false connection, where headlines, visuals, or captions do not support the content, and manipulated content, where genuine information or imagery is manipulated to deceive. Furthermore, some recent frameworks introduced a three-way classification approach. For example, Yang et al. [59] classified news into real, fake, and unverified categories, reflecting the practical ambiguity found in real-time social media streams.

3.5. Discussion on datasets

Multimodal fake news datasets have changed from small-scale manually curated benchmarks to massive automatically collected repositories. This growth introduces a critical trade-off between the data volume and annotation quality. Early datasets, such as LIAR and WEIBO, offer verified labels; however, their limited scale and reliance on simplistic categorical labels often fail to capture the complex spectrum of modern misinformation, including subtle satires or digitally altered contexts. In contrast, massive datasets, including Fakeddit, provide the unprecedented volume required to train deep transformer-based models, but the automated nature of their scraping and labeling processes inevitably introduces noise and potential misalignments between the text and imagery. This quantitative expansion risks compromising the precision of detection systems in high-stakes real-world scenarios, where nuanced judgment is paramount.

Profound domain and cultural biases that restrict the global applicability of current detection frameworks also exist. Most existing benchmarks are heavily centered on US-centric political events or specific global crises, such as the COVID-19 pandemic, which may lead to models that overfit specific linguistic patterns or event-specific cues. This concentration is exacerbated by the predominance of English-language content, which overlooks the unique syntactical, idiomatic, and cultural nuances inherent to

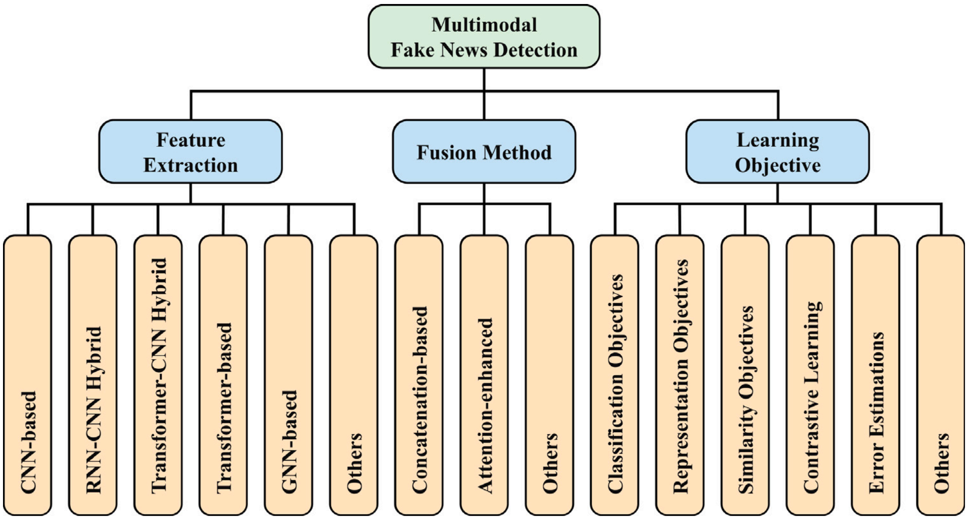


Fig. 4. The structured overview of the model categories. Feature extraction categories are described in Section 4.1, fusion method categories in Section 4.2, and learning objective categories in Section 4.3.

non-Western contexts [60]. Consequently, models validated solely on these benchmarks often struggle with generalizability when they are deployed in diverse linguistic environments. Addressing these biases requires a paradigm shift toward multilingual and culturally diverse data curation, which can only be achieved through collaborative efforts with regional experts and native speakers to ensure that the annotations remain contextually accurate and representative of local disinformation strategies.

The rapid shift in social media consumption highlights a significant modality gap in the current research. While text and image pairs were the primary focus, the rise of short-form video platforms and generative AI necessitates an urgent move toward more complex formats, including audio, video, and interactive synthetic media. Current resources, such as DFDC and AraFacts, begun to address these challenges; however, there remains a substantial need for datasets that provide synchronized, multi-platform, and multi-format data to keep pace with changing deceptive tactics. Ultimately, the research community should prioritize dynamic benchmarking and the integration of emerging AI-generated content to build more robust detection systems that operate reliably across shifting digital landscapes.

4. Multimodal detection

In this section, we analyze multimodal fake news detection models by focusing on three key elements: feature extraction, fusion methods, and the learning objectives. Feature extraction is presented in Section 4.1, fusion methods are discussed in Section 4.2, and the learning objectives are explored in Section 4.3. Figs. 4–6 present a structured overview of the model categories, encoders for each data modality, and fusion methods, respectively. We consider the Web of Science database to select articles. The applied search string is '*(Machine Learning OR Deep Learning) AND (Multimodal) AND (Fake News) AND (Detect*)*'. Among the results, we only consider papers written in English and exclude review papers. The final included papers are summarized in Table 3.

4.1. Feature extraction

The rise of fake news across digital platforms sparks extensive research on multimodal fake news detection. This growing interest is driven by the need to analyze and integrate diverse data modalities, such as text, images, and videos, to accurately identify deceptive content. Multimodal fake news detection harnesses advancements in deep learning and natural language processing (NLP), leveraging state-of-the-art architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformer-based models such as BERT [116], and vision transformers (ViTs) [117]. These approaches aim to capitalize on the complementary strengths of various modalities, offering a more holistic perspective on detecting fake news. In this section, we explore the diverse methodologies employed in multimodal models, categorizing them by their architectural designs and highlighting their unique contributions to this rapidly evolving field.

4.1.1. CNN-based models

CNN-based architectures play a crucial role in the early stages of multimodal fake news detection by enabling effective feature extraction from textual and visual data. The majority of these studies proposed frameworks that integrate text-CNNs with pre-trained image CNNs to exploit the complementary nature of these modalities.

One of the pioneering approaches in multimodal fake news detection, Wang et al. [16], introduced the EANN model, which utilized a text-CNN to extract semantic features from textual data and a pre-trained VGG-19 [118] for visual feature extraction. A

Table 3
Summary of reviewed approaches; *T*: Text, *I*: Image, *A*: Accuracy, *P*: Precision, *R*: Recall, *F*: F1-Score.

Approach	Modality	Datasets	Evaluation metrics
Qian et al. [61]	T, I	WEIBO, Twitter, PHEME	A, P, R, F
Xiong et al. [62]	T, I	WEIBO, GossipCop	A, P, R, F
Qian et al. [63]	T, I, Knowledge	WEIBO, Twitter, PHEME	A, P, R, F
Wei et al. [64]	T, I	WEIBO, Twitter	A, P, R, F
Raj and Meel [65]	T, I	TI-CNN, Emergent, MICC	A, P, R, F
Luqman et al. [66]	T, I	Fakeddit	A, P, R, F, Matthew's correlation coefficient, Odds ratio
Singhal et al. [67]	T, I	Twitter, Facebook	A, P, R, F
Wang et al. [16]	T, I	WEIBO, Twitter	A, P, R, F
Zhang et al. [68]	T, I	WEIBO, Twitter	A, P, R, F
Al-Alshaqi et al. [69]	T, I	ISOT, MediaEval	A, P, R, F
Athira et al. [70]	T, I	FakeNewsNet	A
Segura-Bedmar and Alonso-Bartolome [58]	T, I	Fakeddit	A, P, R, F
Lv et al. [71]	T, I	WEIBO, Twitter	A, P, R, F
Nakamura et al. [47]	T, I	Fakeddit	A
Yang et al. [59]	T, I	WEIBO, Twitter, IKCEST-2023	A, P, R, F
Sharma and Garg [49]	T, I	IFND	A, P, R
Xu et al. [72]	T, I	WEIBO, Twitter	A, P, R, F
Peng et al. [73]	T, I	WEIBO, Twitter	A, P, R, F
Yadav and Gupta [74]	T, I	Twitter, Fakeddit, Jruvika Fake News, Pontes Fake News, and Risdal Fake News Datasets	A, P, R, F
Hua et al. [75]	T, I	ReCOVery	P, R, F
Li et al. [76]	T, I, OCR	BIENdata, ICT and BAAI Dataset	A, P, R
Sharma et al. [77]	T, I	WEIBO, Fakeddit, MediaEval	A, P, R, F
Lu and Ye [78]	T, I, User-side information	WEIBO	A, P, R, F
Huang et al. [79]	T, I	WEIBO, Fakeddit	A, F
Lin et al. [80]	T, I	Fakeddit, GossipCop	A, P, R, F
Liang [81]	T, I	WEIBO, GossipCop	A, P, R, F
Gôlo et al. [82]	T, Linguistic features	Fake.BR, Fact-Checking News, FakeNewsNet, FakeNewsCorpus	F, AUC-ROC
Kumari and Ekbal [83]	T, I	Twitter, WEIBO	A, P, R, F
Yan et al. [84]	T, I, N-HIN	PHEME, GossipCop, MR2	A, P, R, F
Li et al. [85]	T, I	Twitter, WEIBO	A, P, R, F
Guo [86]	T, I	WEIBO, Twitter	A, P, R, F
Al Obaid et al. [87]	T, I	PolitiFact, GossipCop	A, R, P, F, Specificity, G-Mean
Meel and Vishwakarma [44]	T, I	Jruvika Fake News Dataset, All Data, Pontes Fake News Dataset	A, P, R, F
Rani and Shokeen [88]	T, I	PHEME, CrisisLex, ISOT	A, P, R
Guo et al. [89]	T, I, OCR	WEIBO, Twitter	A, P, R, F
Nadeem et al. [90]	T, I	Jruvika Fake News Dataset, TI-CNN, Pontes Fake News Dataset	A, P, R, F
Qu et al. [91]	T, I	GossipCop, PolitiFact	A, R, P, F, TNR
Qu et al. [92]	T, I, Knowledge, Time	GRFN, IFND	A, P, R, F, MCC
Wang et al. [93]	T, I	PolitiFact, GossipCop, Twitter	A, R, P, F
Yadav et al. [94]	T, I	Twitter, Jruvika Fake News, Pontes Fake News, and Risdal Fake News Datasets	A, P, R, F, AUC-ROC
Yan et al. [95]	T, I	ReCOVery, GossipCop, MR2	A, P, R, F, MCC
Ayetiran and Özgöbek [96]	T, I, OCR	PolitiFact, MMHS150K, MultiOFF	A, P, R, F
Nadeem et al. [97]	T, I	PolitiFact, GossipCop	A, P, R, F
Palani et al. [98]	T, I	PolitiFact, GossipCop	A, P, R, F
Wang et al. [99]	T, I	WEIBO, Twitter	A, F
Yu et al. [100]	T, I	WEIBO, Twitter, PolitiFact, GossipCop	A, P, R, F
Wang et al. [101]	T, I, User-side information	WEIBO	A, P, R, F, AUC, AP
Yin et al. [102]	T, I	Twitter, WEIBO	A, P, R, F
Meel and Vishwakarma [103]	T, I	MediaEval, WEIBO	A, P, R, F
Jarrahi and Safari [104]	T, Social context	FakeNewsNet	A, P, R, F
Zhang et al. [105]	T, I	WEIBO, Twitter, Facebook	TP, FP, TN, FN, Gini coefficient
Sciucca et al. [106]	T, I	Fakeddit	A, F

(*continued on next page*)

Table 3 (*continued*).

Approach	Modality	Datasets	Evaluation metrics
Raj and Meel [107]	T, I	CovID, ReCOVery, CoAID, MediaEval	A, P, R, FP, Specificity, Mathew's correction coefficient
Peng et al. [19]	T, I, Audio	Twitter, WEIBO	A, P, R, F
Goyal et al. [108]	T, I, Social context	Twitter	A, P, R, F
Ahuja and Kumar [20]	T, I, Social context	FakeNewsNet, MKLab-ITI	A, F, Sensitivity, Specificity
Kumari et al. [109]	T, Video, Audio	FakeClips, FVCv2.0, FVCv3.0, VAVD	A, P, R, F
Kumari et al. [110]	T, I, Emotion	NovEmoFake	A, F
Azri et al. [111]	T, I, Sentiment	FakeNewsNet, DAT@Z20	A, P, R, F
Kopev et al. [112]	T, Audio, Metadata	CT-FCC-18	MAE, A, F, R
Zhang et al. [113]	T, I, Knowledge	FakeNewsNet, Evons, Fakeddit, Cosmos, MM-Covid, GoodNews, Twitter, PHEME	A, P, R, F
Zhang et al. [114]	T, I, Propagation information	PolitiFact, GossipCop, PHEME	A, P, R, F
Zhang et al. [115]	T, I, Propagation information	PolitiFact, GossipCop, PHEME	A, P, R, F

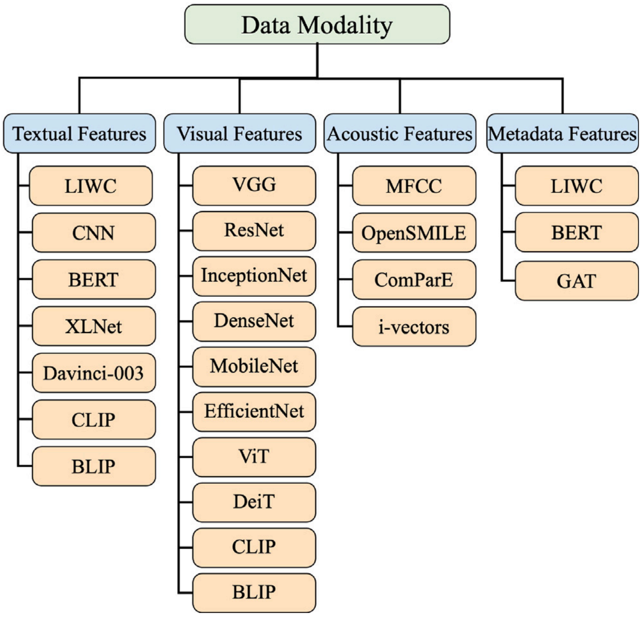


Fig. 5. Encoder of each modality.

distinctive aspect of their approach is the incorporation of adversarial learning, allowing the model to generalize beyond specific news events and effectively identify fake news across diverse contexts. This highlights the potential of CNNs for both text and image modalities and lays the groundwork for integrating adversarial techniques into multimodal architectures.

Building on this foundation, subsequent works explored additional CNN architectures for multimodal fake news detection. For example, Wang et al. [99] employed a similar setup, using a text-CNN for textual features and a pretrained VGG-19 for visual analysis, with an emphasis on effective feature fusion to enhance cross-modal learning. Furthermore, Raj and Meel [65] expanded the scope by evaluating a range of pre-trained CNN models, including ResNet50 [119], VGG-16, InceptionV3 [120], DenseNet [121], and MobileNet [122], combined with a text-CNN for extracting textual features. Their results provide valuable insights into the performance trade-offs between various architectures, offering practical guidance on model selection based on computational resources and dataset characteristics.

Despite the success of CNN-based models in multimodal fake news detection, they exhibit certain limitations, particularly in capturing sequential dependencies within textual data. To address these challenges, several research explored hybrid approaches that combine the sequential modeling capabilities of RNNs with the feature extraction strengths of CNNs. These RNN-CNN-based models aim to bridge this gap by leveraging the complementary strengths of both architectures.

4.1.2. RNN-CNN hybrid models

To address the limitations of CNNs in capturing sequential dependencies, hybrid models that combine RNNs and CNNs are widely adopted for multimodal fake news detection. These RNN-CNN-based models leverage the contextual learning capabilities of RNNs for text data and the feature extraction power of CNNs for images, enabling a more comprehensive multimodal analysis.

Kumari and Ekbal [83] introduced an attention-enhanced framework that employed Bi-directional Long Short-Term Memory (Bi-LSTM) [123] with attention mechanisms for textual data, combined with CNN and Bi-directional Gated Recurrent Unit (Bi-GRU) [124] for visual features. Similarly, Raj and Meel [107], Sharma and Garg [49], and Guo [86] used LSTMs to extract sequential text features and pre-trained VGG networks for visual feature extraction. Moreover, Guo [86] demonstrated the importance of low-level visual features such as textures and edges for effective fake news detection. Additionally, Ayetiran and Özgöbek [96] proposed a novel approach that processed not only text and image data but also the relationships between images and text. They utilized tools such as LAVIS [125] for generating image captions and EasyOCR for extracting text embedded in images. Bi-LSTM and CNN layers were then applied to generate feature vectors for these multimodal inputs, demonstrating the potential of integrating textual, visual, and contextual features for enhanced fake news detection.

4.1.3. Transformer-CNN hybrid models

Unlike fully transformer-based models, Transformer-CNN hybrid models retain CNN encoders for visual feature extraction while adopting transformer-based architectures, particularly BERT, for textual encoding. This hybrid design bridges the gap between sequential modeling and contextual understanding, addressing the limitations of RNN-CNN approaches in capturing global context across long textual sequences.

Among Transformer-based models, BERT emerged as a notable innovation. Its bidirectional self-attention mechanism allows for a more effective contextual understanding of the text, addressing the limitations of capturing context and nuanced dependencies across long text sequences. This capability, combined with pre-training on large-scale corpora and fine-tuning for specific downstream tasks, enabled BERT to achieve state-of-the-art performance in various domains, including fake news detection and multimodal analysis.

Following this trend, numerous studies on multimodal fake news detection adopted BERT as a text encoder, leveraging its powerful contextual understanding to enhance performance across both text and image modalities. BERT can be combined with various visual encoders, such as VGG [62,73,93], ResNet [61,69,80], EfficientNet [77], and Inception-ResNet-v2 [103]. Representatively, Nakamura et al. [47] introduced the Fakeddit dataset [126], which evaluated multiple pre-trained text models (BERT, InferSent [127]) and image models (VGG-16, ResNet50, EfficientNet-b4 [128]), providing a valuable foundation for future multimodal studies. However, to overcome the limitation of pooling operations, which ignore spatial relationships in images, Palani et al. [98] employed BERT and CapsNet [129] to extract textual and visual features, respectively. CapsNet's ability to capture spatial hierarchies in images allowed for a more accurate representation of visual content in the multimodal framework.

Subsequent research [64,71,75] enhanced Transformer-CNN architectures by introducing innovative components. Wei et al. [64] employed BERT and VGG-19 for feature extraction and introduced novel loss functions, such as reconstruction loss, modality discriminator loss, and event discriminator loss, to improve cross-modal representation learning. Hua et al. [75] utilized a back-translation method to augment text data, generating additional text data by translating the original text and then translating it back. They incorporated both the original and back-translated texts, along with the corresponding images, to train the model. Text and image features were extracted using BERT and ResNet50, enabling effective multimodal fake news detection. Furthermore, attention mechanisms within transformers were integrated into hybrid architectures to enhance fake news detection. For example, Guo et al. [89] extracted embedded text from images using PP-OCRv3 [130] and used them as auxiliary features in a multimodal model.

Advancing to BERT-based approaches, several research adopted XLNet [131], which extends BERT using permutation-based training, allowing it to capture bidirectional context more effectively. Al Obaid et al. [87] built an ensemble of attention-based models using XLNet, Bi-GRU, and VGG to aggregate predictions. Qu et al. [91] proposed a novel quantum multimodal fusion-based model, combining quantum encoding and quantum CNNs. They summarized the textual content using text-davinci-003 and embedded it using a pre-trained XLNet. Image features were extracted using VGG. Each modality feature is concatenated and processed using quantum circuits. Amplitude encoding encodes data into quantum states. Three-layer quantum CNNs extract discriminative features from the qubits.

4.1.4. Transformer-based models

In contrast to hybrid approaches aforementioned, fully transformer-based models adopt transformer architectures for both textual and visual modalities. Representative examples include ViT for image patch tokenization and cross-modal pre-trained models such as CLIP and BLIP, which learn joint text-image representations within a unified embedding space.

Huang et al. [79] and Yadav et al. [94] employed ViT for image feature extraction and BERT for textual feature representation, emphasizing the complementary synergy between these two architectures in processing multimodal inputs. Similarly, Yu et al. [100] extended this framework by integrating BERT with Data-efficient Image Transformers (DeiT) [132], a lightweight version of ViT, which enhanced the model's ability to capture both contextual and visual embeddings with improved computational efficiency.

Building on these architectural advances, recent studies increasingly adopted Contrastive Language-Image Pre-training (CLIP) frameworks [85], marking a significant leap in multimodal learning. In the context of multimodal fake news detection, Xu et al. [72] demonstrated the efficacy of pretrained CLIP models in extracting comprehensive visual-linguistic features. Liang [81] extended this approach by incorporating Bootstrapping Language-Image Pre-training (BLIP) [133], further enhancing multimodal feature extraction. Yan et al. [95] proposed a hybrid framework combining CLIP with BERT and ResNet architectures to optimize both textual and visual feature extraction. Similarly, Zhang et al. [105] implemented an advanced approach combining RoBERTa for textual analysis with CLIP-ViT for visual processing, facilitating robust cross-modal feature fusion. These Transformer-based frameworks represent a significant advancement in multimodal fake news detection, providing robust feature extraction that captures the contextual and semantic nuances of both text and images. However, challenges such as efficient large-scale data processing and effective cross-modal alignment remain, underscoring the need for advanced architectures that can seamlessly integrate multimodal features while maintaining their computational efficiency and cross-modal performance.

4.1.5. GNN-based models

The propagation of fake news often occurs through complex social media networks, where the relationships among users, posts, and interactions (e.g., comments, likes, and shares) play a critical role in the dissemination process [134]. Graph Neural Networks (GNNs) are particularly effective for fake news detection because of their ability to model relational data and capture dependencies within graph-structured information. By representing posts, users, and social interactions as nodes and edges in a graph, GNNs provide a comprehensive understanding of the structural and contextual patterns underlying the propagation of fake news. Furthermore, GNNs' capacity to integrate multimodal and auxiliary information makes them invaluable for addressing the challenges of fake news detection.

Incorporating social context, such as followers and likes, significantly enhances the effectiveness of GNN-based models for detecting fake news. Goyal et al. [108], Yan et al. [84] and Ahuja and Kumar [20] demonstrated the integration of textual, visual, and social features within graph-based frameworks. Goyal et al. [108] processed textual data using LSTMs to extract linguistic and contextual patterns, while visual data, including profile and banner images, were analyzed using CNN architectures. They further incorporated network-based features, such as the number of follows, likes, retweets, and mentions, by encoding these relationships using Graph Convolutional Networks (GCNs) to model the social dynamics of fake news dissemination. Similarly, Ahuja and Kumar [20] expanded the scope by integrating three modalities—textual, visual, and social context—using BERT, VGG, and GNNs [135] to extract and fuse features across modalities. These studies collectively highlight the versatility of GNNs in integrating multimodal and social information to provide a robust framework for detecting fake news. Lastly, Yan et al. [84] constructed a heterogeneous information network that integrated news posts, user profiles, and comments to semantically align textual and visual representations. Their work demonstrated how GNNs effectively capture the relationships between different modalities and social interactions, which are crucial for detecting fake news in complex online environments.

In addition to incorporating social context, GNNs are used to extract features for fake news detection. For example, Qian et al. [63] introduced a knowledge conceptualization module to extract entities from text data. These entities were linked to external knowledge graphs to create conceptual explanations, which, when combined with the text, generated graph-structured data. A Graph Convolutional Network (GCN) [136] was then applied to extract textual features, while image features were processed using a VGG network. Similarly, Qu et al. [92] integrated temporal and content information by clustering news with similar topics and constructing graphs of entities extracted from news texts and images using BERT and a scene graph generator, respectively. Graph Attention Networks (GATs) [137] and GCNs were then employed to generate multimodal and temporal features, emphasizing the importance of leveraging both graph structures and temporal relationships in understanding fake news dissemination patterns.

4.1.6. Others

In multimodal fake news detection, several studies adopted additional features, such as sentiment or emotion, user-specific characteristics, and audio, demonstrating that these features can significantly enhance detection performance. These features offer complementary perspectives to traditional textual and visual modalities, providing nuanced insights into the psychological, social, and auditory dimensions of information dissemination. For example, a series of sentiment and emotion analyses can capture the underlying tone and intent of the content, which can often reveal manipulation or deceptive patterns [138,139]. This section reviews research efforts that incorporate these features, highlighting their contributions to the advancement of multimodal fake news detection systems.

Several approaches were proposed to integrate sentiment and emotion features for fake news detection. In sentiment analysis, rule-based methods such as VADER [111] and TextBlob [66] are commonly employed to detect fake news. Other studies depended on pre-trained deep learning models for more nuanced and robust emotion feature extraction. Wang et al. [101] used sentiment scores derived from SKEP [140], a pre-trained sentiment analysis tool. Kumari et al. [110] categorized images into six emotion classes (joy, love, sadness, fear, surprise, and anger) and extracted emotion features using ResNet. Similarly, Yadav and Gupta [74] extracted emotional features from images using a pre-trained VGG network and from text using emoticons combined with the NRC word-emo

User-oriented features were also utilized in multimodal fake news detection. Lu and Ye [78] extracted text features using BERT and obtained contextual features from posts, such as user profile descriptions and follower/following counts. Several other studies incorporated user-oriented features, including like counts, follower counts, and user activity metrics [101,104].

In addition to textual, visual, and user-oriented features, recent studies explored audio and video modalities for multimodal fake news detection. Kumari et al. [109] and Peng et al. [19] combined BERT for text encoding, VGG-19 for video feature extraction, and various audio processing techniques, including MFCC features and OpenSMILE, to capture acoustic patterns for sentiment analysis. Similarly, Kopev et al. [112] utilized textual features, such as LIWC categories and BERT embeddings, alongside acoustic features derived from ComParE [141] and *i*-vectors [142], integrating metadata such as speaker IDs for a comprehensive multimodal framework.

Recent advancements in multimodal fake news detection attempted to refine feature extraction and fusion techniques by leveraging domain knowledge and integrating diverse modalities into videos. Choi and Ko [143] proposed a domain-adaptive framework combining textual, acoustic, and user-centric features with video-specific cues, incorporating ResNet and BERT-based embeddings to enhance multimodal feature alignment. Furthermore, Qi et al. [144] and Zeng et al. [145] explored a more comprehensive multimodal approach by simultaneously leveraging textual, visual, acoustic, and video modalities. These studies emphasize the importance of incorporating multiple data sources to improve detection accuracy and mitigate the biases inherent in unimodal or weakly multimodal systems. By integrating diverse modalities, they aim to capture the complex nature of fake news spread across social media and video-sharing platforms, making detection more robust and context-aware.

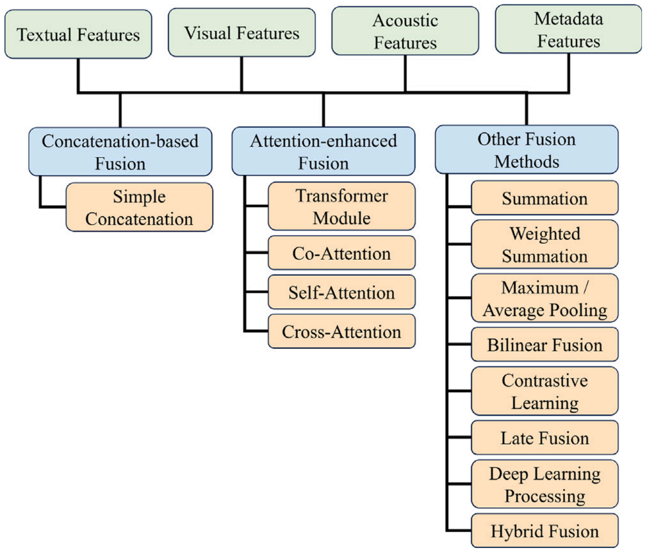


Fig. 6. Diverse fusion methods used to generate multimodal representations explicated in Section 4.2.

In summary, the diverse approaches employed in multimodal fake news detection highlight the potential of leveraging multiple modalities to improve detection performance. CNNs-, RNN-CNN hybrid-, transformer-, and GNNs each demonstrate unique strengths in capturing spatial, temporal, and relational patterns within multimodal data. Moreover, the integration of complementary features, such as sentiment, user-specific metadata, and audio-visual cues, significantly enriches detection frameworks, providing nuanced insights that go beyond traditional text and image analysis. These advancements emphasize the importance of unified and adaptable multimodal approaches for building robust fake news detection systems.

4.1.7. Discussion on feature extraction

Each feature extraction paradigm presents distinct trade-offs in terms of performance, computational cost, and applicability. CNN-based models are lightweight and suitable for real-time scenarios; however, they lack the ability to capture long-range textual dependencies. RNN-CNN hybrids improve sequential text modeling but struggle with global contexts across long documents. Transformer-CNN and fully transformer-based models achieve stronger cross-modal semantic understanding at the cost of higher computational demands, making them less practical in resource-constrained environments than other models. GNN-based models uniquely capture social propagation patterns, offering advantages in detecting coordinated misinformation. However, they require rich social graph data that may not always be available. Overall, the choice of architecture should be guided by the available modalities, computational budget, and the specific nature of the misinformation being targeted.

4.2. Fusion method

4.2.1. Concatenation-based fusion approaches

Several research employed concatenation as a strategy to fuse textual and visual features for multimodal representation. For instance, Wei et al. [64] applied modality and event discriminator losses with feature concatenation to enhance the fusion process. Similarly, Wang et al. [16], Singhal et al. [67], Zhang et al. [68], and Luqman et al. [66] used simple concatenation to integrate image and text features. Segura-Bedmar and Alonso-Bartolome [58] also employed concatenation, but with embeddings extracted via CNN models.

Further studies, such as Sharma and Garg [49] and Sharma et al. [77], and Yadav and Gupta [74], leveraged concatenation to generate robust multimodal features. In addition, Al Obaid et al. [87] and Guo et al. [89] incorporated concatenation with classifier layers to make the final predictions. Kopev et al. [112] significantly highlighted the flexibility of concatenation by integrating features from multiple modalities, including text, image, acoustic, and metadata.

4.2.2. Attention-enhanced fusion techniques

Attention mechanisms, often used in combination with concatenation, are effective for multimodal fusion because they enhance the interaction between textual and visual features. For example, Qian et al. [61] introduced a multimodal contextual attention network that leverages transformer modules to fuse text and image features. Zhang et al. [113] also employed cross attention for the multimodal features. Similarly, Xiong et al. [62] designed an advanced architecture incorporating an Inconsistency Measurement Module that evaluates modality similarity. This is followed by co-attention layers and multiple concatenation operations to achieve

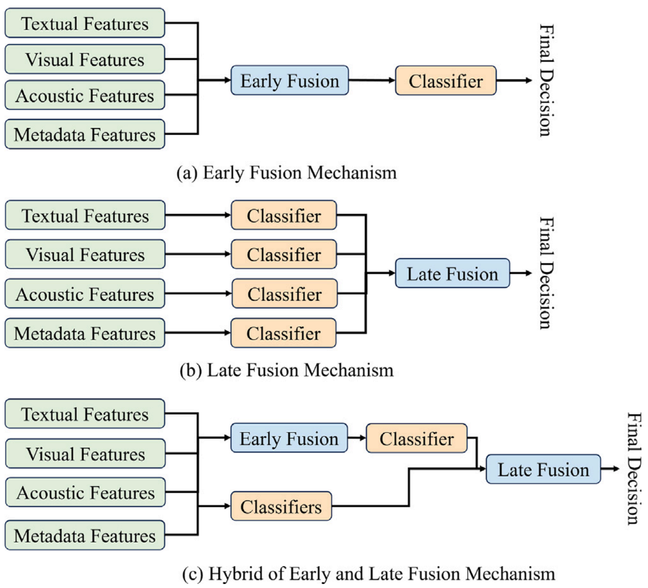


Fig. 7. The different stages of the model in which fusion takes place, discussed in Section 4.2.

a robust multimodal representation. Qian et al. [63] also employed attention mechanisms, applying a co-attention module to GCN and VGG-19 features, followed by pooling and concatenation to finalize feature integration.

In addition, several researchers explored multi-head self-attention for feature fusion. For instance, Hua et al. [75] and Yang et al. [59] utilized self-attention mechanisms to integrate extracted textual and visual features, enhancing modality interactions and generating unified multimodal representations. Peng et al. [73] applied cross multi-head attention to effectively align and fuse text and image embeddings. Similarly, Lu and Ye [78] and Huang et al. [79] leveraged self-attention layers to merge modality-specific features, resulting in improved multimodal fusion performance.

Cross-modal attention emerged as an effective method for feature integration in multimodal fake news detection. Liang [81] employed cross-modal attention to refine modality-specific feature embeddings before their fusion. Similarly, Guo [86] proposed a mutual attention mechanism in which one modality serves as input to the attention mechanism of the other, generating enriched modality features that are then concatenated for the final representation. Further extending these concepts, Yadav et al. [94] introduced visual-semantic and self-attention blocks to extract salient features, ensuring that image and text features complement each other during the fusion process.

Incorporating attention mechanisms into multimodal transformers also yielded promising results. Yan et al. [95] and Zhang et al. [105] utilized co-attention and cross-fusion mechanisms, respectively, to achieve tight alignment between textual and visual modalities. Ayetiran and Özgöbek [96] combined inter-modal attention with weighted feature concatenation, emphasizing the most important features from each modality. Lv et al. [71] further refined attention-based techniques by employing cross-attention on unimodal embeddings, followed by late fusion, to ensure that modality-specific contributions were well represented in the final output. Similarly, Zhang et al. [115] combined self-attention and cross-attention to integrate the temporal and structural features of propagation paths, enhancing the early detection of fake news from multimodal inputs. These studies demonstrate the adaptability of attention mechanisms, enabling deeper and more meaningful interactions between modalities.

4.2.3. Others

Some fusion approaches combine multiple fusion techniques to capture the intricate relationships among modalities. Nakamura et al. [47] explored a variety of fusion methods, including summation, concatenation, maximum pooling, and average pooling, to integrate image and text embeddings, demonstrating the versatility of combining different strategies. Similarly, Xu et al. [72] employed bilinear fusion alongside attention modules to generate robust multimodal features, highlighting the strength of blending simple and advanced fusion techniques for improved feature integration. Zhang et al. [114] combined attention mechanism and diffusion modeling to fuse features of various modalities and constructed propagation graph to detect fake news propagation path.

Contrastive learning is considered a powerful tool in feature fusion frameworks. Kumari et al. [109] applied contrastive learning to align modality-specific features within a shared latent space, ensuring coherence between textual and visual representations. Similarly, Kumari et al. [110] used contrastive learning to align multimodal feature representations with target classes, enhancing inter-modal synergy and class discrimination. These approaches illustrate how these strategies improve feature-level understanding and multimodal representation through the use of alignment techniques. Furthermore, Yan et al. [84] adaptively concatenated contrastively learned, semantically aligned, and network-derived features to create a comprehensive representation of multimodal inputs.

Stacking features and modality-specific processing are key components of advanced fusion strategies. Li et al. [85] stacked learned features and fused them with CNN-MobileNet V2 to create a robust post representation. Azri et al. [111] concatenated text and sentiment features using stacked LSTM layers to create a joint representation, which was then fused with visual features to generate a comprehensive multimodal representation. These studies demonstrate the flexibility and effectiveness of combining stacked architectures with advanced fusion techniques to capture complex multimodal relationships.

Sophisticated fusion method architectures seamlessly integrate early, late, and custom fusion techniques. For instance, Peng et al. [19] introduced a hierarchical approach that combines multiple fusion strategies to capture both intermodal and intramodal correlations. This framework incorporates deep metric learning to enhance features, ensuring robust integration across the text, visual, and acoustic modalities. Similarly, Yin et al. [102] employed a multi-step fusion strategy, beginning with early fusion for basic feature alignment, advancing to joint fusion using co-attention transformers, and concluding with late fusion to integrate the enhanced features. These iterative fusion techniques enable a comprehensive understanding of multimodal inputs.

Other advanced techniques involve the weighted combination of different modalities. Yu et al. [100] combined linear adaptation, self-attention, and gate components to fuse textual and visual features. They then concatenated the fused textual and visual features, along with their differences, to form the final integrated features. Similarly, Nadeem et al. [97] introduced feature stacking and weighted summation to integrate the unimodal and multimodal predictions. Raj and Meel [107] demonstrated the effectiveness of weighted-average fusion by experimenting with various strategies, concluding that this approach outperforms simpler fusion methods.

In conclusion, advanced fusion methods provide flexibility and robustness, making them well suited for a wide range of modalities and tasks. By combining early, late, and joint fusion techniques with attention mechanisms, contrastive learning, and domain-specific knowledge, these methods achieve state-of-the-art multimodal integration performance.

4.2.4. Stages of feature fusion: Early, late, and hybrid fusions

Feature fusion methods in multimodal fake news detection are typically categorized into three types: *early fusion*, *late fusion*, and *hybrid fusion* [146]. This section reviews representative works associated with each fusion strategy and provides the authors' own analysis of their practical implications, trade-offs, and considerations for real-world deployment.

Early fusion. In early fusion, as illustrated in Fig. 7(a), features from different modalities are combined immediately after the feature extraction. Common fusion operations at this stage include concatenation, attention mechanisms, and other vector-level integration techniques such as multi-head self-attention. The resulting multimodal feature representations are then passed to a classifier for prediction.

Late fusion. Late fusion strategies, as shown in Fig. 7(b), involve applying individual classifiers to each modality and then combining the resulting output probabilities. Raj and Meel [65] proposed a weighted late fusion technique, where modality-specific prediction scores are optimized through grid search to improve accuracy. Goyal et al. [108] similarly adopted a weighted summation method to aggregate unimodal outputs, ensuring that each modality contributes appropriately to the final decision. Jarrahi and Safari [104] explored late fusion between latent text features and publisher-related metadata, demonstrating how domain-specific signals can enhance detection performance.

Hybrid fusion. Hybrid fusion combines both early and late fusion stages, as shown in Fig. 7(c). In this approach, features from multiple modalities are first fused through early fusion and then passed to individual classifiers. The resulting predictions are further aggregated using late fusion. Lin et al. [80] investigated this strategy and found that combining joint and decision-level fusion methods improved multimodal performance across various benchmarks.

4.2.5. Discussion on fusion method

Different fusion strategies offer distinct advantages depending on the task requirements, computational budget, and modality availability. Concatenation-based methods remain attractive because of their simplicity, low computational cost, and ease of implementation. These approaches are particularly suitable for real-time fake news detection systems, in which rapid inference is required and all modalities are consistently available. However, simple concatenation often fails to capture the deeper semantic relationships between textual and visual features, which may limit performance in more complex multimodal scenarios.

Attention-based fusion methods provide stronger cross-modal reasoning by explicitly modeling the interactions between modalities. In particular, self-attention, cross-attention, and co-attention mechanisms enable the model to identify salient features and align semantically relevant information across modalities. These methods are particularly effective for detecting fake news involving misleading captions, manipulated images, or subtle inconsistencies between textual and visual content. Nevertheless, these methods typically require substantially higher computational resources and memory consumption than simple fusion techniques, making them less suitable for low-resource or real-time deployment environments.

Similarly, contrastive learning-based fusion methods have shown strong potential for aligning modality-specific representations within a shared embedding space. By encouraging semantically related text-image pairs to be closer while separating unrelated pairs, contrastive learning improves cross-modal consistency and domain generalization. These methods are particularly useful when the goal is to enhance robustness across datasets collected from different domains or social media platforms. However, contrastive learning often requires large-scale training data, carefully designed positive and negative sample pairs, and additional computational overhead during training.

Based on the reviewed literature, each fusion strategy offers unique strengths and tradeoffs. Early fusion methods, such as simple concatenation, are lightweight and computationally efficient, making them suitable for real-time applications, such as fake news detection during live debates. However, although more computationally expensive, complex fusion techniques (e.g., attention-based methods) often deliver superior accuracy owing to their ability to model intricate feature interactions.

Late fusion is particularly advantageous in scenarios in which some modalities may be missing or unreliable. Because classification is performed independently for each modality, late fusion supports the flexible integration of partial or incomplete data, which is an important property in dynamic environments.

Hybrid fusion strategies aim to balance performance and robustness, although they typically involve high computational costs. These methods leverage the strengths of both early and late fusion, offering improved generalization, particularly when domain-specific and contextual features are available.

Despite the promising results reported in the literature, several limitations remain. First, most existing methods assume that all modalities are always available, which is often unrealistic in real-world applications. Second, few studies conduct systematic comparisons among early, late, and hybrid fusion methods under the same experimental settings, making it difficult to identify the most effective strategy for a given application scenario. In addition, fake news detection in general remains challenging under noisy, incomplete, or low-quality multimodal inputs, such as blurred images, short texts, or missing contextual information.

Future research may focus on adaptive fusion frameworks that dynamically assign modality weights according to the reliability and quality of each modality. In addition, there is a growing need for lightweight multimodal fusion methods that support real-time inferences in resource-constrained environments. More robust missing-modality fusion strategies, explainable multimodal architectures, and cross-domain generalization techniques are also likely to become important research directions in the future. Finally, recent advances in large language models and vision-language models suggest that trainable fusion modules combined with pre-trained multimodal encoders may provide a promising direction for improving both robustness and generalization in multimodal fake news detection.

4.3. Learning objective

4.3.1. Classification objectives

To detect fake news, the task can be formulated as a binary classification, where the goal is to classify news as either fake or real. Machine learning approaches, such as Logistic Regression (LR) and support vector machines (SVM), were utilized for this task [80]. For deep learning approaches, the sigmoid function is typically applied at the output layer to map the raw score from the model to a probability $p \in [0, 1]$ [72,76,77]. This formulation is optimized using binary cross-entropy loss [65,97,102], defined as follows:

$$\mathcal{L}_{BCE} = -(y \log(p) + (1 - y) \log(1 - p)), \tag{1}$$

where y represents the ground truth label. In contrast, when the task requires identifying multiple specific categories of misinformation, the softmax function is utilized to generate a probability distribution across K classes [70,81,98]. This approach is paired with the categorical cross-entropy loss [63,71,94,100], which minimizes the divergence between the predicted distribution and the one-hot encoded ground truth:

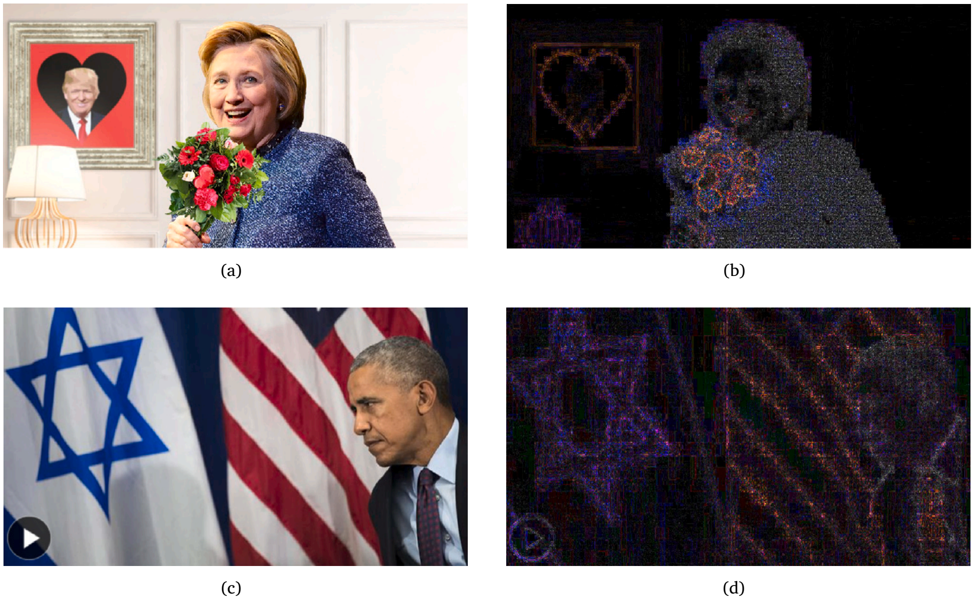
$$\mathcal{L}_{CE} = -\sum_{i=1}^{K} y_i \log(\hat{y}_i). \tag{2}$$

4.3.2. Representation objectives

Several approaches were proposed to learn better representations for fake news detection. Wei et al. [64] developed a method that learns representations by reconstructing text and image embeddings while simultaneously classifying news as fake or real. Gôlo et al. [82] introduced a multimodal variational autoencoder that reconstructs text-related metrics, achieving more comprehensive multimodal representations for fake news detection. Another approach leverages multitask learning by incorporating networks that classify event categories of news content [16,68,85,86]. By training the model to minimize event-specific performance, event-related information is excluded, enhancing the model's ability to detect fake news across different events or domains. Kumari et al. [109] emphasized the role of emotional appeal in the dissemination of fake news, integrating audio-visual and textual emotion recognition to capture subtle cues that enrich the model's representations for better fake news detection.

4.3.3. Similarity objectives

Measuring the similarity among data is crucial for detecting fake news. Meel and Vishwakarma [44] calculated cosine similarity between image captions and news bodies, arguing that fake news involving mismatched text and contextually irrelevant images can be identified by analyzing their coherence. Similar approaches can be applied to measure the similarity between text, images, and OCR text [89], as well as between news body summaries, images, and news headlines [90]. Cosine similarity is also utilized in loss functions to align representations with identical labels [84]. Moreover, techniques such as triplet learning [19,73,79] and Chebyshev distance [93] are mainly employed as loss functions for representation alignment.



(a) (b) (c) (d)

Fig. 8. Examples of ELA. (a) and (c) are forged images, (b) and (d) are ELA-processed images.

4.3.4. Contrastive learning

Contrastive learning is an effective method for generating embeddings of pairs that share similar information while distancing those that have different content. Researchers used contrastive learning to cluster the representations of fake news by utilizing the labels of training samples, ensuring that fake news content is closely represented [19,75,95,110]. Another approach to contrastive learning involves pairing representations from different modalities. Yan et al. [84] paired textual and visual information to position embeddings from the same content close together. Similarly, Kumari et al. [109] employed contrastive learning using text, audio and video embeddings. Beyond modality-level, contrastive learning has also been applied to social graphs to capture structural and semantic relations. For instance, Zhang et al. [115] proposed a dual-view self-supervised contrastive learning framework to jointly model the global propagation topology and high-order heterogeneous relationships among users, publishers, and news content. By integrating node-level and meta-path-based contrastive objectives, their method yields enriched user embeddings, which are subsequently used for reinforced propagation path generation and early-stage fake news detection.

4.3.5. Error estimations

Meel and Vishwakarma [44] estimated noise inconsistencies using K-means clustering on image blocks, asserting that manipulated images exhibit varying noise levels depending on the location within the image. Error Level Analysis (ELA) [147] was employed to assess the extremeness of error levels for fake news detection [44,90]. The ELA is a forensic technique used to determine whether an image was digitally altered. Fig. 8 illustrates examples of ELA.

4.3.6. Others

Wang et al. [99] adopted few-shot learning to enhance fake news detection for emerging events. They utilized meta-learning to efficiently adapt the detector to new events. Al Obaid et al. [87] highlighted the significant class imbalance in real-world news, where real news vastly outnumbers fake news. To address this, they employed focal loss, which is more suitable for imbalanced datasets than cross-entropy loss. In addition, Gôlo et al. [82] used one-class learning and trained their model solely on real news. Qu et al. [92] proposed a two-level cross-entropy loss approach that incorporates temporal information, arguing that consecutive news on the same topic offers complementary insights. Finally, Zhang et al. [105] developed Bayesian classifiers based on Bayes' theorem, leveraging their ability to generalize effectively.

4.3.7. Discussion on learning objective

The evolution of learning objectives in multimodal fake news detection reflects a transition from simple label-based classification to structure-aware representation learning. While classification objectives remain the standard for final decision-making, they often lack the granularity to capture nuanced deceptive patterns between modalities. Binary cross-entropy is computationally efficient but can fail to address the inherent semantic gap between textual and visual data if used in isolation.

In contrast, representation and similarity objectives offer a robust framework by forcing the model to learn the underlying coherence between news bodies and images. These approaches are particularly effective in detecting out-of-context fake news, where the individual modalities are authentic, but their combination is misleading. However, these objectives increase architectural complexity and require carefully curated data to avoid learning trivial correlations.

Contrastive learning aligns high-dimensional features across modalities by maximizing the agreement between related pairs. This enhances the generalizability of the model across different domains and news events. However, the performance of contrastive objectives is highly sensitive to the quality and diversity of negative samples, which poses a challenge for small-scale datasets. Furthermore, as noted in error estimations, learning to identify digital manipulation using forensic-driven objectives adds a critical layer of defense that purely semantic models might overlook.

Finally, addressing real-world data imbalance remains a pivotal challenge. Although focal loss and meta-learning offer potential solutions, their integration into multimodal pipelines can cause training instability. Future research should prioritize the development of hybrid learning objectives that can simultaneously optimize for classification accuracy, cross-modal alignment, and robustness against emerging AI-generated content.

5. Future research

Despite significant advancements in multimodal fake news detection, several challenges persist. By addressing these directions, the field can advance toward creating robust, versatile, and impactful solutions for combating fake news in an ever-evolving digital landscape.

5.1. Cross-platform detection

Most existing studies concentrated on data obtained from a single platform, which limits the generalizability of detection models across different environments. Future research should focus on developing models that can detect fake news across multiple platforms. Each platform presents unique modalities and characteristics, and addressing the absent modalities is key to improving robustness. Approaches such as conditional generative frameworks [148] and cross-modal transformers [149] can be explored to handle incomplete data in multimodal settings, offering a pathway to more adaptable models.

5.2. AI-generated content

As artificial intelligence continues to advance, new challenges in fake news detection are emerging. The evolution of generative models, such as ChatGPT, introduces AI-generated fake news, which exhibits distinct characteristics compared with human-crafted misinformation [150]. These developments underscore the need to create advanced detection models capable of addressing not only traditional fake news but also the unique attributes of AI-generated content. Future research should prioritize the integration of multimodal approaches and leverage cutting-edge techniques to build robust systems that can adapt to this rapidly evolving landscape.

5.3. Modalities

As social media increasingly transitions from text- and image-based content to video-based platforms [14,151], detecting fake news in these new formats becomes imperative. Additionally, with the rise of deepfake technologies, the detection of fake news in video and audio formats is becoming increasingly critical. However, recent research focused primarily on text and image modalities. Future studies should prioritize the development of specialized datasets and models for detecting fake news in video and audio formats to address the emerging challenges and opportunities in this field.

5.4. Contextual information

Modern platforms offer a wealth of auxiliary information beyond the primary content of news items, including user interaction patterns, content metadata, and temporal data. Future research should focus on integrating this auxiliary information into fake news detection models by using appropriate fusion methods. By incorporating such data, models can achieve more nuanced and accurate detection, capturing contextual and relational dynamics that extend beyond the content itself. This approach can provide a deeper understanding of the factors influencing news propagation, ultimately improving the robustness and reliability of the detection systems.

5.5. Learning objectives

Most studies mainly relied on classification-centric objectives, such as cross-entropy loss. However, these objectives may not fully address real-world challenges, including handling imbalanced datasets and capturing uncertainty in predictions. Future research should explore alternative learning objectives. Self-supervised learning approaches, such as bootstrapping [152], can enhance the representation of multimodal data. Adversarial training methods can improve robustness by generating and detecting adversarial examples [153,154]. Adversarial domain adaptation can improve generalizability and adaptability [155]. By diversifying the learning objectives, models can better align with practical requirements and perform more effectively in diverse and dynamic real-world environments.

5.6. System-level implementation

Several real-world fact-checking services, such as PolitiFact, address multiple modalities, including images and videos, but they primarily focus on text. These services rely on human-driven verification processes to ensure accuracy and a contextual understanding. However, this human-centric approach inherently limits scalability, making it challenging to keep pace with the growing volume and rapid dissemination of information in the field. This highlights a critical bottleneck in effectively addressing the speed and scope of the propagation of fake news. Consequently, the development and deployment of automated systems are essential to address these challenges and enhance the efficiency and scalability of fact-checking processes. Although most prior research focused on developing detection models, understanding how these models can be effectively deployed in real-world scenarios is equally critical. For instance, Rani and Shokeen [88] proposed a fake news detection system that combines deep learning and blockchain technology to enhance reliability and transparency, highlighting a promising pathway for practical applications. However, research on the practical deployment and integration of detection models into operational systems remains limited. Future studies should explore scalable and efficient system architectures that consider computational resources tailored to diverse contexts. Moreover, addressing ethical concerns, including user privacy and data security, is essential for fostering societal trust and broader adoption, specifically when handling user data such as user metadata or interaction data. Furthermore, such systems must be designed to operate without introducing biases or generating fake news. Transparent mechanisms that provide explainable and interpretable outputs are vital for ensuring accountability and user confidence in such systems.

6. Conclusion

The rapid advancement of digital media and social networks has established fake news as a critical societal challenge that undermines public trust and influences political decisions. While early detection efforts relied on unimodal data, such as text or images, they often fall short of addressing the complex nature of fake news, which frequently combines multiple modalities to create more persuasive and deceptive narratives. In response, multimodal approaches, which integrate and analyze information from various sources, including text, images, videos, and metadata, have emerged as promising and essential solutions.

This study systematically examines the advancements in multimodal fake news detection, providing a comprehensive review of the field and outlining key directions for future research. Specifically, we propose a novel taxonomy that structures multimodal approaches along three critical dimensions: feature extraction, fusion methods, and learning objectives. This framework allows for an in-depth analysis of existing models and a detailed comparison of key datasets, highlighting their characteristics, limitations, and applicability to multimodal tasks.

We trace the evolution of feature extraction from early CNN-based models to more advanced transformer-based architectures (e.g., BERT, ViT) and Graph Neural Networks (GNNs), which better capture contextual and relational data. We analyze the progression of fusion methods from simple concatenation to sophisticated attention-enhanced and hybrid strategies that model complex inter-modal relationships. Finally, we examine the diversity of learning objectives employed, moving beyond standard classification to include representation learning, similarity objectives, and contrastive learning, which enables the construction of more robust models.

This comprehensive analysis reveals several critical challenges, including the lack of diverse datasets that limit model generalizability and the new frontier of detecting AI-generated content, that is, deepfakes. Addressing these hurdles requires future work to focus on integrating auxiliary information, such as user interaction patterns and metadata, to build context-aware systems. Furthermore, exploring alternative learning objectives, such as self-supervised and adversarial training, is key to enhancing model robustness. Ultimately, transitioning these advanced models from research to real-world applications is the final, critical step, requiring scalable, transparent, and ethical systems. This necessitates a collaborative effort between academia, industry, and policymakers to foster a resilient and trustworthy digital-information ecosystem.

CRediT authorship contribution statement

Juyeob Lee: Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Junyeop Cha:** Writing – original draft, Supervision, Software, Resources, Investigation, Conceptualization. **Minsu Park:** Writing – original draft, Methodology, Data curation, Conceptualization. **Minyoung Lee:** Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Taeeun Kim:** Visualization, Supervision, Resources, Funding acquisition, Formal analysis, Data curation. **Seul-Ki Choi:** Visualization, Software, Resources, Funding acquisition, Data curation. **Angel P. del Pobil:** Visualization, Supervision, Funding acquisition. **Eunil Park:** Writing – original draft, Visualization, Validation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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